

Solution Requirements

Date	15 March 2026
Team ID	XXXXXX
Project Name	EcoTrack – AI-Powered Carbon Footprint Tracker
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	Authentication	Users can securely register and log in using Firebase Authentication.	High
2	Authorization Levels	Authenticated users can manage only their own carbon footprint data, history, and profile.	High
3	External Interfaces	Integrates Firebase Firestore, Carbon Interface API, Groq AI API (LLaMA 3.3-70B), and Chart.js.	High
4	Transactions Processing	Processes travel, food, and energy inputs, calculates carbon emissions, stores results, and updates the dashboard.	High
5	Reporting	Generates carbon footprint history, trend charts, heatmaps, carbon budget progress, badges, and downloadable reports.	High
6	Business Rules	Calculates emissions using verified emission factors, applies food waste penalties, tracks streaks, and awards badges based on sustainable activities.	Medium
7	Compliance to Laws or Regulations	Protects user data using Firebase security rules and follows privacy and secure authentication practices.	High
8	Other	Generates AI-powered eco-friendly recommendations based on daily carbon emission patterns using Groq AI.	High

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	Dashboard, AI suggestions, and carbon calculations should load quickly.	Page response time < 2 seconds
2	Scalability	System should support increasing numbers of users and carbon records.	Supports thousands of users and cloud data storage
3	Security & Data Privacy	Secure authentication, encrypted communication, and protected cloud database access.	Firebase Authentication with HTTPS encryption
4	Reliability & Availability	System should remain available with minimal downtime.	99.9% uptime
5	Usability & Accessibility	Responsive interface that works across desktop, tablet, and mobile devices with easy navigation.	Mobile-friendly and responsive UI
6	Other	Maintainable modular architecture using React.js, Flask, and Firebase for future enhancements.	Easy maintenance and extensibility